# IRC Results Snapshot

Refined GAD → IRC validation, status as of 2026-04-15

## What This Note Covers

This note summarizes the *current* IRC validation workflow and the latest scientifically meaningful refined rerun. It is intentionally compact: the goal is to preserve the key numbers, the threshold analysis, and a small amount of interpretation in one place.

**Pipeline used here**

1. Start from converged GAD trajectories from `noise_survey_300`.
2. These trajectories come from **dataset TS geometries** with Gaussian noise added to the `Transition1x` labeled transition-state coordinates.
3. Select a TS candidate from the saved trajectory (`best_nneg1`).
4. Recompute projected TS quality with HIP: $n_{\text{neg}}$, force norm, and $f_{\max}$.
5. If needed, refine the TS candidate with projected GAD.
6. Run IRC *from that TS* in both downhill directions using Sella's IRC routine.
7. Compare the two endpoints against the `Transition1x` labeled reactant/product using: aligned geometry (element-aware Kabsch/Hungarian RMSD) and bond-topology matching.

**Important interpretation point** The current labels `intended / half / unintended` mean: *does the HIP IRC from this GAD-found TS recover the dataset-labeled reactant/product pair?* This is a strong benchmark, but not a complete unsupervised analysis of all possible connected minima on HIP's PES.

## Run Summary

The refined rerun analyzed here is the completed array `59385481_[0-2]`. It covered 30 converged TS candidates each at noise levels 0, 10, and 50 pm.

| Noise | Total | IRC ran | Gate failed | Intended | Half | Topology-half only / Unintended |
|---|---|---|---|---|---|---|
| 0 pm | 30 | 15 | 15 | 1 | 3 | 1 / 10 |
| 10 pm | 30 | 7 | 23 | 0 | 2 | 1 / 4 |
| 50 pm | 30 | 3 | 27 | 0 | 0 | 1 / 2 |

**Headline:** the dominant failure mode is now `ts_quality_gate_failed`, not code failure. The looser refined gate did recover one fully intended case at 0 pm and increased IRC attempts there (15 instead of 12), but the attempted cases still become progressively worse with noise: a few mixed / half-intended outcomes at low noise and mostly unintended behavior by 50 pm.

# Main Figure

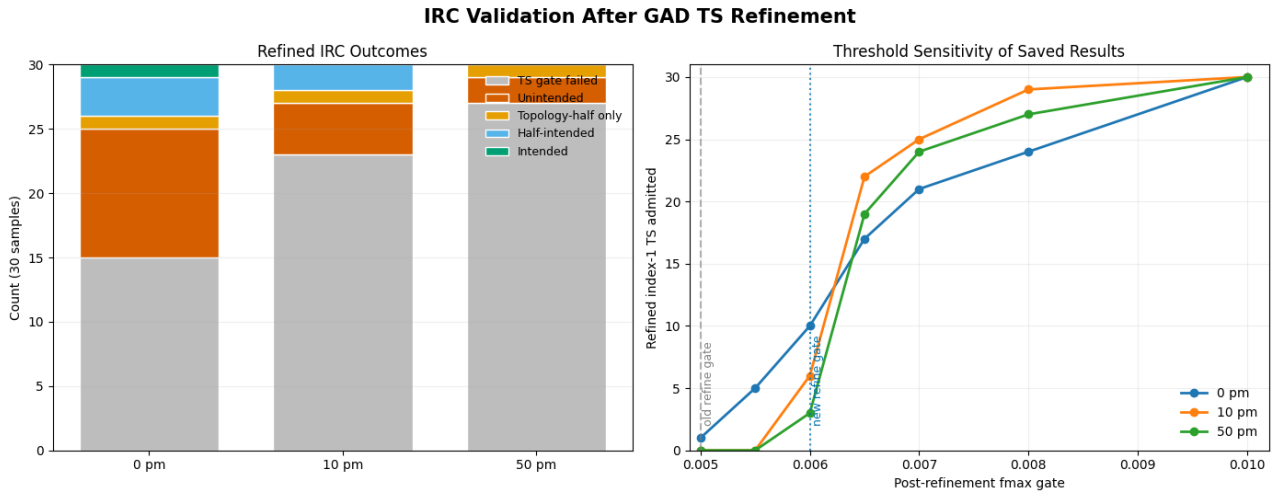


**Figure 1: Left:** outcome breakdown of the refined rerun. **Right:** number of refined index-1 candidates that would pass under looser post-refinement $f_{\max}$ gates, using the already-saved results.

### Interpretation

- Noise hurts in two layers: fewer candidates survive TS screening, and the surviving ones connect less cleanly to the labeled endpoints.
- Even 0 pm is only partially "clean" under the stricter pipeline: one intended case, a few half-intended cases, and many gate failures. This means the old convergence notion ($n_{\text{neg}} = 1$ plus loose force screening) was broad enough that some nominally converged points were not clean TSs by stricter $f_{\max}$-based standards.
- The threshold-sweep panel shows why moving from $f_{\max} < 0.005$ to $f_{\max} < 0.006$ was justified. The current rerun already reflects that change.

# Why The Force Gate Was Too Strict

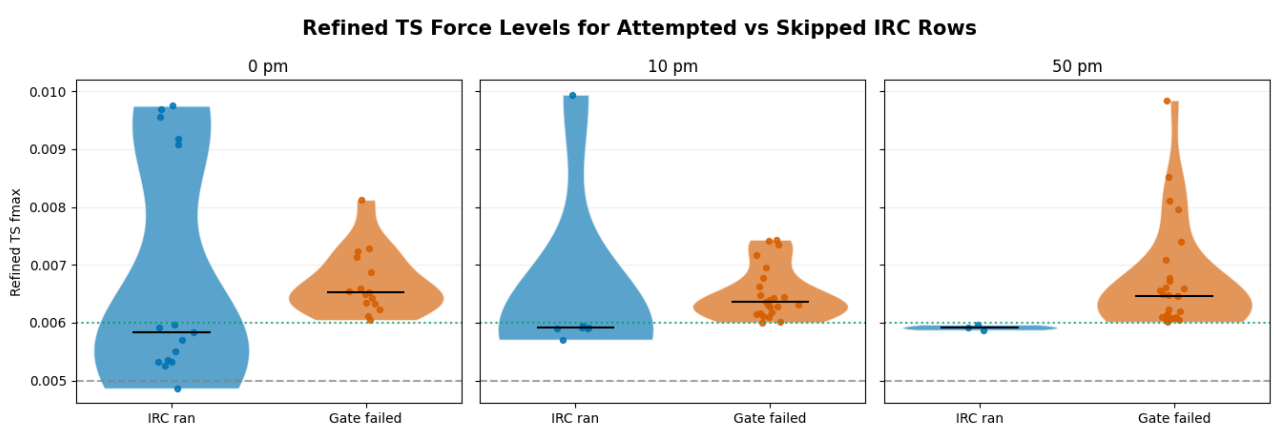


**Figure 2:** Refined TS $f_{\max}$ for rows that actually ran IRC versus rows that were skipped by the TS-quality gate. Dashed gray: old refined gate 0.005. Dotted green: new refined gate 0.006.

**Most important observation:** the previous policy had been asymmetric, and the current rerun is

the first one after correcting that.

- Unrefined candidates could go straight to IRC if they satisfied $f_{\max} < 0.01$.
- Refined candidates were only allowed through if they satisfied $f_{\max} < 0.005$.

That old policy meant some *improved* refined TSs were being rejected even though their $f_{\max}$ was smaller than already-accepted unrefined TSs. The current rerun used the corrected post-refinement gate $f_{\max} < 0.006$.

| Noise | Max accepted pre-gate $f_{\max}$ | Min rejected refined $f_{\max}$ | Median rejected refined $f_{\max}$ |
|---|---|---|---|
| 0 pm | 0.009748 | 0.006202 | 0.006533 |
| 10 pm | 0.009941 | 0.006007 | 0.006357 |
| 50 pm | — | 0.006018 | 0.006456 |

**Conclusion:**

- 0.005 was stricter than necessary for a first refined IRC pass.
- The current rerun confirms that $f_{\max} < 0.006$ is a reasonable next operating point: it increases attempted IRC at 0 pm and recovers one intended case without simply reverting to the original 0.01 screen.
- The remaining failed-gate rows are now clustered around 0.0062–0.0065, which is tighter and more interpretable than the old 0.0052–0.0059 bottleneck.

## Updated Rerun Settings

The new defaults already patched into the repo are:

- `--refine-steps 600` instead of 300
- `--refine-force-threshold 0.006` instead of 0.005

Why 600 steps? In the completed refined rerun, the refinement stage still occasionally hit the new 600-step cap, especially at 50 pm:

| Noise | Rows hitting 600-step refinement cap | Mean refinement steps |
|---|---|---|
| 0 pm | 2 | 87 |
| 10 pm | 0 | 127 |
| 50 pm | 3 | 201 |

So the main gain from the latest rerun came from the looser refined gate rather than dramatically longer refinement trajectories; most cases did not use the full 600-step budget.

## What To Inspect First In 3D

The viewer bundles from the completed refined run are already available under: `/lustre07/scratch/memoozd/gadpl`

Recommended cases:

- `0 pm / sample 31`: the one fully intended case in the latest rerun
- `0 pm / samples 0, 33, 45`: representative low-noise half-intended cases
- `10 pm / samples 51, 175`: useful geometry-vs-topology disagreement cases
- `50 pm / sample 175`: topology-half at high noise

- `50 pm / samples 20, 89`: clearly unintended high-noise examples

Each bundle contains:

- dataset reactant reference
- reverse IRC endpoint
- TS input
- forward IRC endpoint
- dataset product reference

So this is an endpoint/context view of the IRC outcome, not a dense movie of every internal IRC integration step.

## Bottom Line

- The current refined IRC pipeline is now producing *real* scientific signal.
- The main issue is TS admissibility, not numerical IRC crashes.
- Noise degrades both TS quality and endpoint agreement.
- The latest rerun under $f_{\max} < 0.006$ is better at 0 pm, but high-noise cases remain dominated by gate failures and unintended endpoints.
- The next decision point is not "whether to loosen from 0.005" anymore; that is already done. The next question is whether further loosening or a different TS-selection/refinement strategy is justified.

## What Is Running Right Now

### Queue snapshot captured at 2026-04-15 17:22 UTC

| JobID | Name | State | Time | Node / Reason |
|---|---|---|---|---|
| 59385266_1 | run_nr_then_gad.slur | R | 1:17:47 | ng20601 |
| 59385266_2 | run_nr_then_gad.slur | R | 1:17:47 | ng20601 |
| 59385266_3 | run_nr_then_gad.slur | R | 1:17:47 | ng30102 |
| 59385266_4 | run_nr_then_gad.slur | R | 1:17:47 | ng30102 |
| 59385266_5 | run_nr_then_gad.slur | R | 1:17:47 | ng30102 |
| 59385266_7 | run_nr_then_gad.slur | R | 1:17:47 | ng20401 |
| 59385266_8 | run_nr_then_gad.slur | R | 1:17:47 | ng20401 |
| 59385266_9 | run_nr_then_gad.slur | R | 1:17:47 | ng20401 |
| 59385266_10 | run_nr_then_gad.slur | R | 1:14:18 | ng30104 |
| 59385266_11 | run_nr_then_gad.slur | R | 1:14:18 | ng20401 |

The latest IRC rerun `59385481_[0-2]` has already completed. The jobs still running now are the `run_nr_then_gad` experiments shown above.